Example 2

Let $S = \{-3\alpha, -\alpha, \alpha, 3\alpha\}$; $A = \{-3\alpha, \alpha\}$ and $B = \{\alpha\}$

Are A and B independent given $P(-3\alpha) = P(-\alpha) = 1/8$ $P(\alpha) = 1/4$ $P(3\alpha) = 1/2$

Soln

$A = \{3\alpha, \alpha\}$

$B = \{\alpha\}$

$A \cap B = \{\alpha\}$

$P(A) = 1/8 + 1/4 = 3/8$

$P(A \cap B) = 1/4$

$P(B) = 1/4$

for independent

$P(A \cap B) = P(A)\,P(B)$

$= 3/8 \times 1/4 = 3/32$

$P(A \cap B) \neq P(A \cap B)$

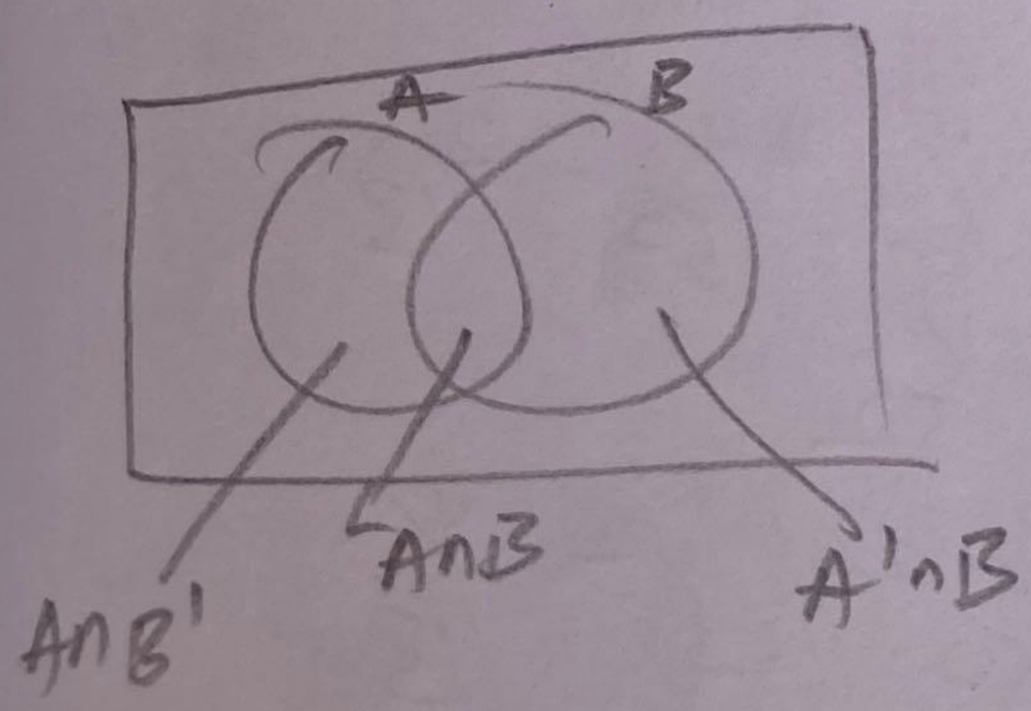
$1/4 \neq 3/32$

A and B are not dependent

Example 3

If A and B are independent show that A and $\bar{B}$ are also independent

Soln



For independent

$P(A \cap B) = P(A)\,P(B)$

$A = A \cap B' + A \cap B$

$P(A) = P(A \cap B') + P(A \cap B)$

$P(A \cap B') = P(A) - P(A \cap B)$

$P(A \cap B') = P(A) - P(A)P(B)$

$P(A \cap B') = P(A)(1 - P(B))$

$P(B') = 1 - P(B)$

$P(A \cap B') = P(A)\,P(B')$

Example 4

Let $S = \{-3\alpha, -\alpha, \alpha, 3\alpha\}$ $P(-3\alpha) = P(-\alpha) = 1/8$ $P(\alpha) = 1/4$ $P(3\alpha) = 1/2$

Consider a block of 10 symbols.

Solns

a. Prob of all symbols belong to block A

$A = \{-3\alpha, \alpha\}$

$P(A) = (1/8 + 1/4)^{10} = (3/8)^{10} = 5.499 \times 10^{-5}$

b. Prob of none symbols belong to block A

$\bar{A} = \{3\alpha, -\alpha\}$

$P(\bar{A}) = (1/2 + 1/8)^{10} = (5/8)^{10} = 9.095 \times 10^{-3}$

c. Prob of at least one symbol belong to block A

$1 - P(\bar{A}) = 1 - 9.095 \times 10^{-3} = 0.991$

Example 5

5a. $P(T) = 1/5$ $P(Pl) = 2/3$

$P(T \text{ or } Pl) = P(T \cup Pl) = P(T) + P(Pl) = 1/5 + 2/3 = 13/15$

5b. $P(Ace) = 4/52$ $P(King) = 4/52$

$P(Ace \text{ or } King) = P(A \cup K) = P(A) + P(K) = 4/52 + 4/52 = 2/13$

5c. $P(A) = 4/52$ $P(S) = 13/52$ $P(A \cap S) = 1/52$

$P(A \cup S) = P(A) + P(S) - P(A \cap S) = 4/52 + 13/52 - 1/52 = 4/13$

LECTURE 2

Example 1

$n(S) = {}^5C_2 = 10$

$P(X=2) = 3/10$ $P(X=1) = 6/10$ $P(X=0) = 1/10$

Example 2

$P(X=0) = {}^3C_0/8 = 1/8$

$n(S) = {}^3C_2 + {}^3C_1 + {}^3C_0 + {}^3C_3 = 8$

$P(X=3) = {}^3C_3/8 = 1/8$ $P(X=2) = {}^3C_2/8 = 3/8$ $P(X=1)\ {}^3C_1/8 = 1/8$ $P(X=0) =$

Example 3

$n(S) = \begin{matrix} HH & TH \\ HT & TT \end{matrix}$ or 2^n where n = nuber of tuss

or you count $= 2^3 = 8$

$n(S) = 8$

$P(X=0) = 1/8$ $P(X=1) = 2/8 = 1/4$ $P(X=2) = 1/8$

Example 4

$n(S) = (H, T)$ Prob is either a 0 or 1

$I = \begin{cases} 0 & P(I=1) = 1 \\ 1 & P(I=0) = 0 \end{cases}$

Example 5

$P(X \geq 3) = 0.20 + 0.15 + 0.10 + 0.05$

$= 0.50$

$P(X = odd) = P(1) + P(3) + P(5)$

$= 0.20 + 0.20 + 0.10$

$= 0.50$

Example 6

HHH HHT HTH
THH THT HTT
TTH TTT

(i) $n(S) = 8$ $P(X=0) = 1/8$ $P(X=1) = 3/8 = 1/4$

$P(X=3) = 1/8$ $P(X=2) = 3/8$

(ii) $P(Y=0) = 5/8$ $P(Y=1) = 3/8$

$F(Y) = \begin{cases} 5/8 & Y=0 \\ 1 & Y=1 \end{cases}$

$F(X) = \begin{cases} 1/8 & X=0 \\ 3/8 & X=1 \\ & X=2 \end{cases}$

$F(X) = \begin{cases} 1/8 & X=0 \\ 4/8 & X=1 \\ 7/8 & X=2 \\ 1 & X=3 \end{cases}$

Example 7

i) $0 + a + 2a + 2a + 3a + a^2 + 2a^2 + 7a^2 + a = 1$

$10a^2 + 9a - 1 = 0$

$a = 1/10$ or -1

$\therefore a = 0.1$ or $1/10$

(ii) $P(X<6) = a^2 + 3a + 2a + 2a + a + 0$

$= 0.1^2 + 3(0.1) + 2(0.1) + 2(0.1) + 0.1 + 0$

$= 0.81$

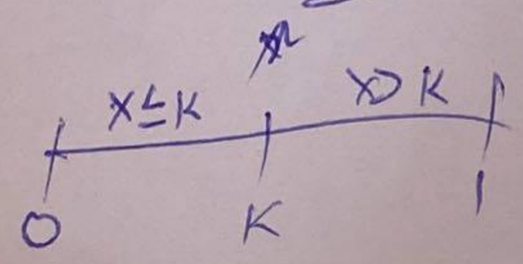
$P(X \geq 6) = 1 - P(X<6)$

$= 1 - 0.81$

$= 0.19$

$$F(x) = \begin{cases} 0 & x=0 \\ 0.1 & x=1 \\ 0.3 & x=2 \\ 0.5 & x=3 \\ 0.8 & x=4 \\ 0.81 & x=5 \\ 0.83 & x=6 \\ 1 & x=7 \end{cases}$$

Example 8



since $x \leq k = x > k$

$x + x = 1 \quad \therefore x \leq k = 1/2$

$2x = 1 \quad x > k = 1/2$

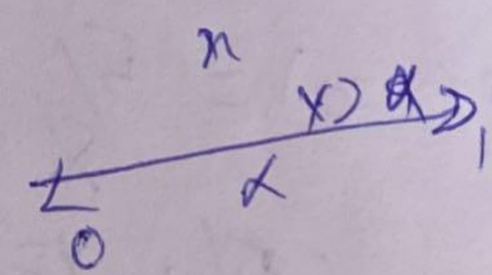
$2 = 1/2$

$\int_0^k 3x^2\,dx = 1/2 \Rightarrow k^3 - 0^3 = 1/2$

$\left[\frac{x^3}{1}\right]_0^k = 1/2 \quad \therefore \quad k^3 = 1/2$

$k = \left[\frac{1}{2}\right]^{1/3}$

$\cancel{k^3 = 1/3 = 1/2}$



$\int_\alpha^1 3x^2\,dx = 0.1$

$\left[x^3\right]_\alpha^1 = 0.1$

$1^3 - \alpha^3 = 0.1$

$\alpha^3 = 1 - 0.1$

$\alpha = [0.9]^{1/3}$

Example 9

$$f(t) = \int_0^{100} 0\,dx + \int_{100}^{150} \frac{100}{x^2}\,dx \qquad \frac{x^{-2+1}}{-2+1} = \frac{x^{-1}}{-1}$$

$$= 0 + 100\left[-\frac{1}{x}\right]_{100}^{150}$$

$$= 100\left[\frac{-1}{150} + \frac{1}{100}\right]$$

$$= 1/3$$

$$\therefore P(x) = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$$

(ii) $\cancel{\left(1 - \frac{1}{3}\right)^3}$ $P(x^+) = 1 - P(x)$

$$= \left[1 - \frac{1}{3}\right]^3 = 8/27$$

Example 10

$$E(X) = \frac{1}{6}(1+2+3+4+5+6)$$

$$= 3.5$$

$$E(X^2) = \frac{1}{6}(1^2+2^2+3^2+4^2+5^2+6^2)$$

$$= 15.17$$

$$Var(X) = E(X^2) - (E(X))^2$$

$$= 15.17 - 3.5^2$$

$$= 2.92$$

Example 11

Money won $= \frac{1}{6}(20+40-30+0)$ Ghc

$$= \frac{1}{6}(30) = 5 \text{ Ghc}$$

Example 12

$$E(X) = \frac{1}{6}(3) + \frac{1}{2}(6) + \frac{1}{3}(9)$$

$$= 6.5$$

$$E(X^2) = \frac{1}{6}(3^2) + \frac{1}{2}(6^2) + \frac{1}{3}(9^2)$$

$$= 46.5$$

$$E[(1+2X)^2]$$

$$E[1 + \cancel{4}X + 4X^2]$$

$$\cancel{E[1]} + E[1] + 4E[X] + 4E[X^2]$$

$$= 1 + 4(6.5) + 4(46.5)$$

$$= 213$$